Introduction
Conditionals in Everyday Life and in Philosophy

A conditional sentence typically has a main clause and at least one if-clause. Often there is just one if-clause. For example:

> If it were done when 'tis done, then 'twere well it were done quickly.
> If our newspaper was delivered this morning, the neighbours' dog took it.
> If happiness is activity in accordance with virtue, it is reasonable that it should be in accordance with the highest virtue.

Conditional sentences with several if-clauses are common.

> If you quit smoking, and if you don't also change your eating and drinking habits, you will probably gain weight.
>
> Yet, they, should the last scene be there,
> The great stage curtain about to drop,
> If worthy their prominent part in the play,
> Do not break up their lines to weep.

Conditional sentences have received concentrated, if intermittent, theoretical attention since antiquity. The attention during the past forty years has been intense. I shall presently sketch the main developments, early and late, and also discuss how conditionals play central roles in many philosophical theories. In the first part of these introductory remarks, I show how someone with no knowledge of the history of the treatment of conditionals and no prior acquaintance with philosophical topics that lead to

an interest in conditionals can still see how conditionals pose a theoretical or philosophical problem. Conditionals play a central role in our everyday thinking about the world and in our deliberating about how we should try to act. One need not have other philosophic concerns to appreciate their importance. One can become puzzled by conditionals without first developing a taste for philosophic puzzles.

Many creatures make their ways in the world without consciously considering any possible alternatives for action. Highly sophisticated and intelligent human activities, for example, rapid, clever conversation, occur unaccompanied by any prior conscious laying of plans about how best to act. People who ponder at length before acting can end up opting for a stupid and disastrous alternative. The practice of deliberation nevertheless continues to enjoy good reputation, and not merely because it appears to be uniquely human. Many projects and accomplishments would be impossible without it.

Deliberation involves the conscious consideration of alternatives for future action. A student entering college and contemplating a career in medicine might consider all the following:

> If I want to be admitted to a good medical school, I have to get decent grades in physics and organic chemistry.
> If I take physics, I need to take calculus first.
> If I go to medical school, I will have to borrow lots of money.
> On the other hand, I can end up earning much more than a teacher or a musician if I become a doctor.
> If I become a doctor, I will frequently have to go without much sleep, especially during my residency.

The consideration of conditionals like these is essential to deliberation. It will be useful to adopt some terminology for talking about the parts of a conditional sentence.

In a conditional sentence such as:

> If you become a doctor, you will make a lot of money,

the first clause, *you become a doctor*, is often called the *antecedent*, and the second clause, *you will make a lot of money*,

is called the *consequent*. Occasionally the old-fashioned grammatical terms *protasis* and *apodosis* are used instead of *antecedent* and *consequent* respectively. With this particular example, and many others, *antecedent* seems to be an appropriate term in two different ways. The antecedent precedes the consequent in the conditional sentence. Moreover, if the sentence is true, what the antecedent is about, your becoming a doctor, will be or would be prior to what the consequent is about, your making a lot of money. If you become a doctor, the sentence suggests, you will become a doctor before making a lot of money. You will make a lot of money as a consequence of practising medicine. The consequent is thus as well-named as the antecedent.

The well-entrenched terms *antecedent* and *consequent* are not likely to be replaced soon in the teaching of elementary logic. They nevertheless can hinder theoretical understanding of conditionals. In the first place, the order of the so-called antecendent and consequent is often reversed in conditional sentences. The order of the clauses in our example can be changed without producing any change in meaning:

You will make a lot of money if you become a doctor.

The antecedent is the clause in the scope of *if*, not necessarily the clause that begins the sentence. In the second place, the terms suggest a thesis that is sometimes explicitly advanced, that a real relation of antecedence or consequence corresponds to the antecedent and consequent clauses of a true conditional sentence.[1] What the consequent describes depends on, is a consequence of, what the antecedent describes. What the antecedent describes comes first. A look at some examples, however, reveals that the order of priority can be reversed:

If I take physics, I need to take calculus first.
If the digital clock display blinks, then the power has been interrupted.
If the annual growth ring of a tree is narrower than normal, the rainfall that year was less than normal.

In the second half of this work, I will devote more attention to

the different patterns of dependence that can relate what is described by the clauses of a conditional sentence. The terms *antecedent* and *consequent* are appropriate only to one of several patterns of dependence. These terms have been common in logic and philosophy books for some time, and they occur frequently in this book as well; but I also use some alternative terms that are shorter, more accurate, and less theoretically loaded, namely *if-clause* and *main clause*. An if-clause is a clause in the scope of *if* no matter where it occurs in a conditional sentence. A main clause is one not in the scope of *if*, one that can stand by itself. When one deliberates whether to do so-and-so, one tries to decide, among other things, what will result, what might result, what cannot result, what must come before, and what cannot come before, if one does so-and-so.

A conditional can bear on future action, however, even though neither its if-clause nor its main clause concerns something that is potentially within the agent's control:

> If a hurricane hits this beach, as one did in 1955, most of these houses will be destroyed.
> If interest rates go up, the market value of a bond declines.

Even though it is not up to you whether a certain beach house is destroyed by a storm or whether a certain bond declines in value, it may be up to you whether you become the owner of the house or bond.

We use conditionals to decide what to do. We also use conditionals to decide what to believe, whether or not the belief is immediately relevant to any prospective alternative for action. Much of our inference from evidence is naturally cast in conditional form. By observing a certain kind of electric clock, one can infer whether or not there has been an interruption in the power. By observing the cross section of a tree, one can infer something about the relative amounts of rainfall in successive seasons past.

All the examples of conditional sentences we have considered so far have been declarative. Non-declarative conditionals are familiar in everyday life and in literature. Commands, requests, advice, and principles of action are often given by conditional imperatives:

If you see her, say hello.
If your soil is mostly compacted clay, work in some granulated gypsum.
If music be the food of love, play on.
If anyone would sue you and take your coat, let him have your cloak as well.

Exhortations and questions can be posed in a conditional form:

If the neighbours' dog didn't steal our newspaper, who did?
If I forget thee, O Jerusalem, let my right hand forget her cunning.
If Winter comes, can Spring be far behind?

Non-declarative examples such as these are commonly neglected by theoretical treatments of conditionals, and I shall continue this tradition of negligence. A conditional imperative, after all, seems to pose no vexing problem. I am not advising you, for example, simply to work some granulated gypsum into your soil no matter what. My advice is hypothetical rather than categorical. If the specified condition obtains, that is, your soil is mostly compacted clay, then my advice is operative. Notice that I have just used a declarative conditional to explain the function of a non-declarative conditional.

Questions and imperatives can be sound or silly, appropriate or irrelevant, and so forth. They cannot be true or false. Declarative conditionals, however, like declarative sentences generally, are naturally regarded as being either true or false. This immediately poses a puzzle.

Truths correspond to reality. Falsehoods don't. 'The cat is on the mat' is true if and only if the cat is on the mat. There is no apparent problem in understanding what state of affairs must actually obtain for 'The cat is on the mat' to be true so long as it is obvious in the situation which cat and which mat are being referred to. 'Someone let in the cat' is somewhat more complicated since there may be no particular person referred to by 'someone'. It is still clear enough what kind of situation must obtain if 'Someone let in the cat' is true: Either Tod let in the cat, or Lucy let in the cat, or Charles let in the cat, or. . . . Consider now the conditional 'Someone let in the cat if the cat is on the

mat.' What sort of situation or state of affairs makes it true? We know how to draw a picture of a cat on a mat, or a cat not on a mat, and of a mat with no cat on it. How can we draw a picture of a conditional state of affairs: if the cat is on the mat, then such-and-such? Given an event description, an event so described either occurs in a certain vicinity during a certain period, or it does not occur. There is no such thing as the conditional occurrence of an event. Declarative conditional sentences about occurrences are therefore not about conditional occurrences. What are they about? What in the world makes a declarative conditional sentence true?

Theorists who deny that conditionals are literally true or false still have a question to answer about the acceptability of conditionals. We distinguish acceptable from unacceptable conditionals; and if the difference cannot be explained by reference to the distinction between truth and falsehood, it must be explained some other way.

The theoretical problem of accounting for conditionals can thus be motivated independently of prior theoretical concerns. Logic, the philosophy of reasoning, and the philosophy of language, which we have just approached through everyday examples, focus attention on the nature of conditionals. Many philosophical concerns in addition to logic and the philosophy of language, for example, the treatments of causation, freedom, and physical existence, provide additional motivation. Conditional accounts of causation and of freedom occur in consecutive parts of David Hume's *An Inquiry Concerning Human Understanding* (1748). The following passages also illustrate how, in Hume's hands, innocent phrases such as 'that is' and 'or in other words' can effect a startling transition:

> Similar objects are always conjoined with similar. Of this we have experience. Suitably to this experience, therefore, we may define a cause to be *an object, followed by another, and where all the objects similar to the first are followed by objects similar to the second.* Or in other words *where, if the first object had not been, the second never had existed.* (Section VII, 'Of the idea of necessary connexion', Part II)
>
> By liberty, then, we can only mean *a power of acting or not acting, according to the determinations of the will*; that is, if we

> choose to remain at rest, we may; if we choose to move, we also may. Now this hypothetical liberty is universally allowed to belong to every one who is not a prisoner and in chains. (Section VIII, 'Of liberty and necessity', Part I)

Hume is clearly on to something when he introduces these conditionals. The acceptability of a conditional such as 'If Judy tried to jog three miles, then she would – or at least might – have jogged three miles' appears to be necessary for the acceptability of 'Judy could have jogged three miles; she had the power, liberty, and freedom to do it.' Similarly, the acceptability of a conditional such as 'If James had not taken the vitamin C, then his cold symptoms would not have disappeared so quickly' appears to be necessary to the acceptability of 'Taking the vitamin C caused James's cold symptoms to disappear.' Conditional accounts of causation are part of today's philosophical orthodoxy. Although conditional accounts of freedom receive somewhat less attention right at present, they were intensively discussed for much of the century.[2]

Most if not all of the properties we ascribe are dispositional. Solubility, the standard philosophical example, is the disposition to dissolve. If a soluble substance is put in a suitable solute, it dissolves; if an insoluble substance is put in a liquid of the same kind, it does not dissolve. To have a certain property is typically to behave a certain way if the conditions are appropriate. Some philosophers have held that even physical existence is dispositional. A famous statement of the view that came to be known as *phenomenalism* appears in J. S. Mill's *An Examination of Sir William Hamilton's Philosophy* (1865):

> I believe that Calcutta exists, though I do not perceive it, and that it would still exist if every percipient inhabitant were suddenly to leave the place, or be struck dead. But when I analyse the belief, all I find in it is, that were these events to take place, the Permanent Possibility of Sensation which I call Calcutta would still remain; that if I were suddenly transported to the banks of the Hoogly, I should still have the sensations which, if now present, would lead me to affirm that Calcutta exists here and now. (Chapter XI, 'The psychological theory of the belief in an external world')

Mill's permanent possibility of sensation is the abiding truth of conditionals relating circumstances of sensing with the character of the resulting sensation. (Note that these conditionals, and also those involved in conditional accounts of causation or freedom, have false if-clauses.)

We shall survey a number of attempts to explain what makes a conditional acceptable or true, whether or not its if-clause is false. The explanations are confined to three historical periods, each marked by original innovation and rapid development in logic generally, not just in the treatment of conditionals. The ancient period consists roughly of the fourth and third centuries BC. The modern period begins in the second half of the nineteenth century and continues – we should like to think – to the present. Separated from these two epochs by long periods of stagnation and decline in logic and the philosophy of logic, the medieval period contains many important logical discoveries and much subtle discussion related to conditionals.

The problems of studying ancient logic and of studying medieval logic are in one respect very different. On one hand, many ancient logical treatises are irretrievably lost. Our information is based on second, third, or fourth-hand reports written hundreds of years after the original work. Many medieval books and manuscripts on logic, on the other hand, are there waiting to be retrieved. As medieval historians keep making new discoveries, the medieval period of logical vigour and invention keeps being lengthened. In their authoritative and judicious *The Development of Logic*, for example, William and Martha Kneale write that 'after about 1380 there is little of interest to report about medieval logic'.[3] In an excellent but much more specialized book, published just twelve years later, another scholar writes that:

> Generally speaking, nothing of interest to the logician was said after 1550 at the very latest. Indeed, now that I have written this book, I have compiled a large list of logic texts for the period 1550–1650 which I shall be happy never to open again. On the other hand, an enormous amount of interesting work remains to be done for the period 1450–1550.[4]

However long the medieval and modern periods of logical vigour are eventually reckoned, it still seems unlikely that the history of

logic will ever be regarded as the history of continuous development. There really were some Dark Ages; and, with the exception of Leibniz, the great philosophers from Hobbes to Kant have no claims to greatness because of their contributions to logic.

The next two chapters discuss ancient and medieval treatments of conditionals. Without making any attempt at completeness or scholarly originality, they use historical discussion as the occasion to introduce some basic concepts of logic.[5]